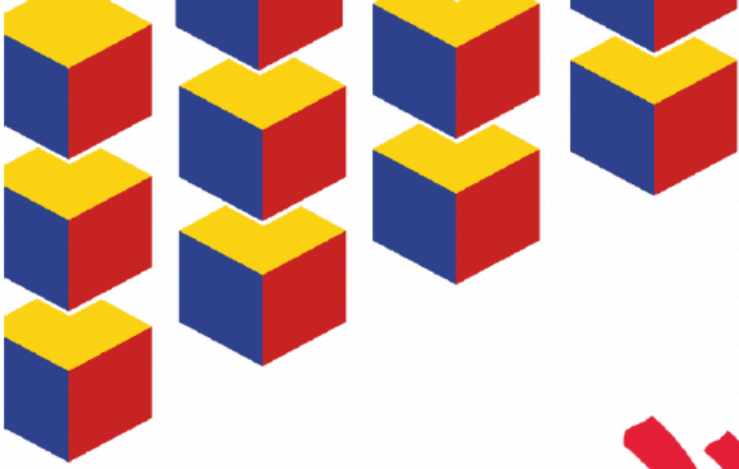




worldskills ASEAN Manila2025

Test Project

Artificial Intelligence (AI)

Structured Data Machine Learning Challenge

Regional World Skills Competition 2026

Artificial Intelligence: Structured Data Machine Learning Challenge

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Introduction

You are working as a Junior Machine Learning Engineer at a leading global technology integration firm. Your engineering division specializes in automated visual inspection, smart workspaces, and interactive Human-Computer Interaction (HCI) software systems.

Due to dynamic requirements from enterprise clients, your division has been tasked with building a lightweight, high-performance computer vision suite utilizing state-of-the-art deep learning architectures. Specifically, you are expected to construct and optimize pipelines for real-time body-part localization (Hand Detection) and dynamic physical attribute identification (Color Profile Detection).

To succeed, you must demonstrate technical agility, robust software engineering practices, and rapid problem-solving capabilities within a strict **9-hour timeframe**.

Workstation & Technical Infrastructure

Competitors are provided with a high-performance deep learning development workstation configured as follows:

- Processor (CPU): Intel Core i7 (12th Generation)
- Graphics Processing Unit (GPU): Dedicated GPU with 16GB VRAM (CUDA-enabled)
- System Memory (RAM): Minimum 16GB / 32GB RAM
- Operating System: Ubuntu Desktop Environment (LTS) / Windows Dual-boot configured with deep learning runtimes.
- Primary IDE: VS Code (Visual Studio Code) is permitted and highly recommended. Jupyter Notebook/JupyterLab environments are also fully supported.
- Programming Language: Python 3.12 - 3.13
- Deep Learning Frameworks: Only YOLOv8 or YOLOv11 architectures are authorized for the detection pipelines.
- Connectivity: Stable Internet access is allowed throughout the competition for pulling pretrained weights, installing packages, checking APIs, and accessing documentation.
- Datasets: The custom evaluation and training datasets will be introduced and delivered to your local storage drive at the exact start time of the competition.

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Project Structure and General Instructions

Folder Naming & Organization

You must structure your workspace strictly to comply with the automated testing runtimes.

Create a root directory named exactly: **WSC2026_C1** on your local Desktop. Inside it, organize your module folders as follows:

None

```
WSC2026_C1/  
├── WSA2026_TP55_**_M1/  
│   ├── left_right.py (or left_right.ipynb)  
│   └── system_documentation_m1.pdf  
└── WSA2026_TP55_**_M2/  
    ├── color_detection.py (or color_detection.ipynb)  
    └── system_documentation_m2.pdf
```

(Replace with your official two-letter country code, e.g., MY, PH, SG, TH, ID, VN, KH).

Key Rules

1. **Time Management:** The entire test package (Module 1 + Module 2 combined) has an absolute limit of 9 hours (540 minutes). You must manage your time carefully. No submissions will be accepted after the 9-hour window.
2. **Execution:** All code must run successfully on the local Intel i7 / 16GB GPU hardware using CUDA acceleration. Code containing runtime failures or syntax errors will receive zero points for execution criteria.
3. **Hardware Constraints:** All physical actuators/external accessories (such as conveyor systems, stepper motors, pilot lamps, and AC LED modules) have been removed. This is a pure software and computer vision detection task. All controls and results must be demonstrated directly via dynamic, real-time bounding boxes, labels, and software interfaces on your screen.

Module 1: Real-Time Left-Right Hand Detection

Description

Develop a real-time deep learning pipeline to detect human hands through a live camera stream and instantly classify them as either a "Left Hand" or a "Right Hand".

System Requirements

Your Python script or notebook (left_right.py or left_right.ipynb) must:

1. Initialize the computer vision feed using the provided camera for detection.
2. Leverage a trained YOLOv8 or YOLOv11 model optimized for high FPS inference on the workstation GPU.
3. Successfully satisfy the following real-time inference logic:
 - a. Left Hand Detected: Detect the hand and output a visual overlay labeling it clearly as "Left Hand" with a corresponding confidence score.
 - b. Right Hand Detected: Detect the hand and output a visual overlay labeling it clearly as "Right Hand" with a corresponding confidence score.
 - c. Both Hands Detected: Detects both hands simultaneously, displaying independent bounding boxes, correct lateral classifications ("Left Hand" and "Right Hand"), and confidence values for each.
4. Render these outputs clearly over the live webcam frame.

Deliverables

Save and organize the following in your WSA2026_TP55__M1 folder:

1. Source Code: left_right.py (Python Script) or left_right.ipynb.
2. Documentation: system_documentation_m1.pdf (Brief technical rundown of your model setup, pipeline diagram, and training parameters).

Module 2: Real-Time Color Detection

Description

Develop a real-time computer vision pipeline to localize target objects and dynamically determine their color profiles from the live video input.

System Requirements

Your Python script or notebook (color_detection.py or color_detection.ipynb) must:

1. Initialize the video frame using the provided camera for detection.
2. Utilize custom YOLOv8 or YOLOv11 object detection models combined with color classification thresholds, or an end-to-end multi-class YOLO model trained on color-specific classes.
3. Successfully satisfy the following real-time requirements:
 - a. Detect targeted objects within the dataset distribution.
 - b. Calculate and extract the primary color profile of each localized object.
 - c. Dynamically project a bounding box and label displaying the Identified Color Name (e.g., "Red", "Blue", "Green", etc.) over the target object in the live viewport.
4. Ensure the color segmentations and labels are mathematically robust to variations in ambient lighting.

Deliverables

Save and organize the following in your WSA2026_TP55__M2 folder:

1. Source Code: color_detection.py (Python Script) or color_detection.ipynb (Jupyter Notebook).
2. Documentation: system_documentation_m2.pdf (Brief technical documentation explaining your color thresholding/color-classification algorithm, model details, and optimization approaches).

Instructions to the Competitor

- **Independent Workspace:** Competitors must work independently and apply the knowledge, skills, and methodologies acquired from professional training. No collaboration, communication, or sharing of resources with other competitors is permitted during the session.
- **Task Scheduling:** The tasks can be completed in any order within the comprehensive 9-hour timeframe. There are no individual module intermediate deadlines; however, you must manage your workflow efficiently to complete and verify both modules before the final session timer expires.
- **File Management & Standard Names:** Ensure you save your work using the correct folder hierarchy and exact filenames specified in Section 3. File submissions placed in wrong locations or structured with custom names will fail automated compilation runs.
- **Internet Permissibility:** Standard internet access is explicitly allowed throughout this competition. Competitors can use the network to download model architectures, pull pre-trained weights from online hubs, refer to documentation, and manage development dependencies.